ACTL3182 Wk9

tadhg

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Recap

Last week, we use Girsanov's Theorem to find a measure $\mathbb{Q}$ that makes the discounted stock price Z_t a $\mathbb{Q}$-martingale; that is,

$$\mathrm{d}Z_t = Z_t \sigma \, \mathrm{d}W_t^{\mathbb{Q}},$$

where

$$\mathrm{d}W_t^{\mathbb{Q}} = \mathrm{d}W_t + \gamma_t \, \mathrm{d}t$$

and

$$\gamma_t = \frac{\mu - r}{\sigma}.$$

Distribution of S_t under $\mathbb{Q}$

Note that

$$\begin{aligned} \mathrm{d}S_t &= S_t(\mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t) \\ &= S_t\left(\mu\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}} - \sigma\frac{\mu - r}{\sigma}\,\mathrm{d}t\right) \\ &= S_t(r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}), \end{aligned}$$

and we showed that this SDE has solution

$$\begin{aligned} S_t &= S_0 \exp\left\{\left(r - \frac{1}{2}\sigma^2\right)t + \sigma W_t^{\mathbb{Q}}\right\} \\ &= \exp\left\{\ln S_0 + \left(r - \frac{1}{2}\sigma^2\right)t + \sigma W_t^{\mathbb{Q}}\right\}. \end{aligned}$$

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Since $W_t^{\mathbb{Q}} \overset{\mathbb{Q}}{\sim} \mathcal{N}(0, t)$, we have that

$$S_t \overset{\mathbb{Q}}{\sim} \mathcal{N}\left(\ln S_0 + \left(r - \frac{1}{2}\sigma^2\right)t, \sigma^2 t\right).$$

Martingale Representation Theorem (continuous)

If Y_t and Z_t are $\mathbb{Q}$-martingales, then there exists a process ϕ_t such that

$$\mathrm{d}Y_t = \phi_t \, \mathrm{d}Z_t.$$

Black Scholes Derivation

Say a derivative pays a random amount X at time T, and let

$$Z_t = S_t/B_t,$$
$$Y_t = \mathbb{E}_{\mathbb{Q}}\left[\frac{1}{B(T)}X \mid \mathcal{F}_t\right]$$

and note both are $\mathbb{Q}$-martingales.

Then, a strategy that holds at time t

- ϕ_t stock
- $\psi_t = Y_t - \phi_t Z_t$ bond

is self-financing and replicating for X.

Black Scholes Derivation

So, the value of the derivative at time 0 must be the value of the portfolio at time 0,

$$\begin{aligned} V_0 &= \phi_0 S_0 + Y_0 - \phi Z_0 \\ &= Y_0 \\ &= e^{-rT}\,\mathbb{E}_{\mathbb{Q}}[X] \end{aligned}$$

Given that under $\mathbb{Q}$,

$$S_t \overset{\mathbb{Q}}{\sim} \mathcal{N}\left(\ln S_0 + \left(r - \frac{1}{2}\sigma^2\right) t, \sigma^2 t\right),$$

this can be used to find derivative prices.

Black Scholes Call

With a call option, we have

$$X = (S_T - K)^+ = (S_T - K)\mathbb{1}(S_T > K).$$

Hence, the price of the call under no arbitrage is

$$\begin{aligned} c_0 &= \mathbb{E}_{\mathbb{Q}}[e^{-rT}(S_T - K)\mathbb{1}(S_T > K)] \\ &= \underbrace{\mathbb{E}_{\mathbb{Q}}[e^{-rT} S_T \mathbb{1}(S_T > K)]}_{=S_0\Phi(d_1)} - Ke^{-rT} \underbrace{\mathbb{E}_{\mathbb{Q}}[\mathbb{1}(S_T > K)]}_{=\Phi(d_2)} \\ &= S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2). \end{aligned}$$

Black Scholes Call Formula Remark

You get given the formula for d_1 and d_2, but you must remember that

$$S_0\Phi(d_1) = \mathbb{E}_{\mathbb{Q}}[e^{-rT}S_T\mathbb{1}(S_T > K)],$$
$$\Phi(d_2) = \mathbb{E}_{\mathbb{Q}}[\mathbb{1}(S_T > K)] = \mathbb{Q}(S_T > K).$$